PY102 · Data Structures & Algorithms in Python
Clock Hours: 50
Quarter Credit Hours: 4
Prerequisites: PY101
Instructors
Dr. David Votaw
DCS (Information Assurance)
Frank Chambergo
John Berta
MS (Classroom Technology)
Neil Kalinowski
MA (Mathematics), MS (Computer Science), MS (Data Analytics)
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MBA
Have an academic question? Submit an Academic Assistance Request, and Student Services will make sure it reaches the right course instructor or member of the academic team.

Course Description
Data Structures & Algorithms in Python introduces students to programming using Python. Students will explore topics such as What Are Data Structures and Algorithms?, Abstract Data Types (ADTs) and Their Role, Efficiency and Performance Basics, Introduction to Big O Notation, Analyzing Time Complexity.

Course Objectives
Data structures and abstract data types
Analyzing algorithm efficiency with Big O notation
Arrays and nested data structures
Implementing stacks and queues
Searching and sorting algorithms
Course Outline
Unit Topics & Assignments
Unit 1 — Foundations of data structures Topics:
• Data organization with arrays and Python lists
• Indexed access, boundary validation, and array insertion
• Array updates, removals, and index shifts
• Iterative list filtering, transformation, and result assembly
• Nested arrays, structural preservation, and safe indexing

Assessment:
Weekly review (quiz)
Unit 2 — Stacks and queues Topics:
• LIFO stacks, push-pop operations, growth, and emptiness
• Stack-based undo behavior and state reversal
• FIFO queues, enqueue-dequeue operations, and stack comparison
• Standard-library imports with import and from-import
• Efficient deque queues and stack-versus-queue selection

Assessment:
Weekly review (quiz)
Unit 3 — Algorithm analysis Topics:
• Algorithm correctness, efficiency, readability, step growth, and user experience
• Big O notation and constant, linear, and quadratic growth
• Big O simplification and worst, best, and average cases
• Complexity of list and dictionary operations
• Stack and queue costs; list-front removal versus deque

Assessment:
Weekly review (quiz)
Unit 4 — Searching and sorting Topics:
• Search processes in everyday situations
• Linear search and worst-case O(n) complexity
• Binary search, sorted data, and O(log n) complexity
• Linear versus binary search scalability
• Bubble, insertion, and Python built-in sorting complexities

Assessment:
Final course review (quiz)
Review format
Reviews in this course consist of a MIX of open-ended and multiple-choice questions,

focused on practical understanding of core concepts: applying key ideas, interpreting code, identifying errors, and reasoning about system behavior.

NOTE: The outline is subject to change at the institution’s discretion

Text & Reference Materials
Proprietary instructional content
Python Docs – Data Structures: https://docs.python.org/3/tutorial/datastructures.html
Teaching Strategies
Individualized tutorials and demonstrations.
LMS Assignments.
Hands-on practice labs.
Projects.
Interactive discussions and case studies.
Grading Policies
Students will be evaluated in the following manner:

Quizzes (Weekly Reviews): 30%.
Online Assignments (Discussions): 10%.
Tests/Exams (Final Review): 60%.
Bonus: Students who submit and pass all reviews on time are eligible to earn up to 10 bonus points.

The grading percentages noted may change at the discretion of the institution.

To satisfactorily complete this course, a student must achieve a minimum of 60 percent (a grade of D). In order to graduate, a student must attain a minimum cumulative GPA of 2.0 and complete all courses with a passing grade.

Grading Scale
Final grades are based on the final grading system found in the catalog.

Grade	Numeric Grades	Grade Points
A	100 – 90	4.0
B	89 – 80	3.0
C	79 – 70	2.0
D	69 – 60	1.0
F	59 & below	0.0

Reviews
All reviews must be completed by the end of the term for course completion. At term end, grades finalize and reviews lock - the only exception is the final review retake, open for two weeks after the last learning day.

Weekly Reviews
A formative assessment that helps students reinforce each unit’s content.

Weekly reviews are scored, but only pass/fail affects your grade: a result of D or higher counts as a Pass and earns full points for that review; a higher passing grade doesn’t add extra course-grade points.
A result below D counts as a Fail and earns no points. A failed attempt does not complete the lesson, and you may try again.
Weekly reviews can be retaken as long as the course remains open for learning.
Weekly reviews are graded automatically, and results are available immediately.
Your first passing attempt completes the lesson and allows you to continue.
Discussions
A weekly question or exercise aligned with the content covered in the current academic unit and the cohort’s standard learning pace.

A discussion question is published each week, and students are encouraged to respond on time and in alignment with the material covered in their current academic unit.
All discussion responses may be submitted until the end of the current course.
When a discussion response is submitted, the final course grade is automatically recalculated and updated based on all grade components.
Final Reviews
The exam is graded by an instructor assigned to the course.
Students may take a retake exam; the higher of the two grades is the final exam grade. The retake window is open for two weeks after the last learning day in the term.
Exam grades will be published within 14 days of submission.
The final course grade will be published on the later of the following dates: eight days

after the course ends, or the date the exam is graded.
If a student takes the retake exam attempt, the final course grade will be updated after that exam is graded.
Instructor Availability
This course requires significant individual effort. If you experience any challenges, contact Student Services as soon as possible for academic support or to connect with your course instructor. Seeking help early can make it easier to stay on track and succeed in the course.

Student Responsibilities
This course is part of Maestro College’s asynchronous distance education program, a non-attendance-taking, academic-event-based program. Coursework is completed through the Maestro Learning Platform in an asynchronous online format, allowing students to access materials at any time throughout the week. There are no scheduled class meetings; however, students are expected to complete assignments at least weekly, meet deadlines, and participate regularly in online discussions and activities.

Attendance & Enrollment
Requirements for attendance in asynchronous distance education programs:

Maestro College Asynchronous Distance Education programs are non-attendance taking, academic event based programs. Although these programs are non-attendance taking, the institution is required to document students’ participation in academic related activity and to terminate the enrollment of any student for which this activity cannot be documented at the following points:

at the end of the first calendar week of the academic term;
at the end of the first four calendar weeks of the academic term;
at the midpoint of each academic term; and
at the end of each academic term.
Example: For a program which is run in 10-week terms a student must complete a qualifying academic activity in weeks 1, 4, 5, and 10 to remain active. Failure to complete a qualifying academic activity during any of these weeks will result in the student being dropped from the program.

Samples of qualifying academic events include:

Attending a synchronous class, lecture, recitation, or field or laboratory activity, physically or online, where there is an opportunity for interaction between the instructor and students
Submitting an academic assignment
Taking an assessment or an exam
Participating in an interactive tutorial, webinar, or other interactive computer-assisted instruction
Participating in a study group, group project, or an online discussion that is assigned by the institution
Interacting with an instructor about academic matters
The following do NOT qualify as academic events:

Logging into an online class or tutorial without any further participation; or
Participating in academic counseling or advisement.
Academic Support
Students enrolled in asynchronous online programs receive academic support through structured, asynchronous support channels aligned with the online instructional model. To request academic assistance, students must submit an academic support ticket through the Maestro Platform, which is administered by Student Services. Academic questions submitted through this system are routed to qualified instructional staff, who provide clarification, guidance, or additional explanation related to course content.

Academic Integrity
You are encouraged to discuss course concepts and assignments with classmates to deepen your understanding, but all submitted work must be your own. Do not copy another person’s work or submit or share solutions that are not your own.

If an academic integrity violation is confirmed, the College may impose sanctions, as determined by the appropriate academic authority and as outlined in the school catalog.

AI Usage
Students may use AI tools for non-graded academic support, such as:

Reviewing material
Asking practice questions
Practicing skills
Students may not:

Use AI to generate graded assignments unless explicitly permitted
Present AI-generated answers as original work without citation
AI misuse may be treated as academic dishonesty.

NOTE: This syllabus is subject to change at the institution’s discretion